

Heredity before individuality in a digital primordial soup

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Theories of the origin of life disagree about what came first but agree about what the transition was: a spreading pattern became an individual, whose future depends on itself rather than on its surroundings¹⁻⁴. That step has never been watched. We watched it in thousands of worlds of random machine code executing one another. Its order is forced by arithmetic: a copier that writes fewer symbols than it reads cannot finish a copy of itself in one pass, so its execution must either run on into its surroundings or come back on itself. The first replicator takes the first way: a two-byte word that writes itself when executed; its offspring take seven of eight bits from the neighbour. Evolution's first act is the second way: one jump that returns execution inside the organism, after which the offspring take nothing from the neighbour. Individuality is a return in control flow, not a boundary in space. In a second machine, one instruction decides whether life begins open and the lethality of the sterile order preceding life decides how the open phase ends. The first living things are the simplest objects their chemistry makes, no more complex than the order around them.

Main

What distinguishes a living thing from the chemistry around it has been answered in two ways. One tradition lists properties, metabolism, compartment and heredity, and organises the origin of life around which came first⁸⁻¹¹. The other locates life in a relation: a living system is one whose operations produce the organisation that produces them, so that its future depends on its own past rather than on its environment, in Maturana and Varela's operational closure¹, Rosen's closure to efficient causation², Fontana and Buss's organisational closure³ and, made quantitative, the information-theoretic individuality of Krakauer and colleagues⁴. The second view coincides with the moment population genetics identifies as the start of Darwinian evolution proper, when vertical descent comes to dominate the communal exchange of an early world^{14,17}. Whatever the first replicating molecules were, this is the transition that made them organisms, and no experimental system has yet allowed it to be observed as it happens, repeated under controlled variation, and taken apart.

Soups of random programs come close. When random byte strings for a simple machine are made to execute one another, with no fitness function and no replacement, self-replicating programs arise and take over⁵, as they did earlier in the Coreworld⁶ and among random Lisp expressions⁷, and as they do for the Z80 processor¹². The phenomenon is robust, and it reproduces the gross signature of a transition to life, a collapse of diversity and a rise in compressibility. What has not been asked is what the first replicator is, why that program and not another, what becomes of it after it fills the soup, and which properties of the machine decide these outcomes; nor has heredity been measured directly, the transition having been inferred from compression statistics of the whole soup. We treated such a soup as an experiment and as a test of one explanation. The explanation is arithmetic: a copier that writes fewer symbols than it reads cannot complete a copy of itself in one pass over its own cells, so its first pass must end either by running on into whatever lies beyond it or by returning to a cell it has already executed. If life in a soup begins with the smallest thing that writes itself, it must begin open, and individuality, in the sense of the theories above, must be a later invention that closes the path of execution. We ran 3,560 worlds of random Z80 code on graphics processors across thirteen deletions of instruction families, program lengths from 3 to 100 bytes, execution budgets, mutation rates and memory geometries, with confirmatory stages pre-registered with predictions and kill criteria (Methods). Heredity is measured by intervention: a program is lifted out of the soup, executed against fresh random partners, and its offspring are tested in turn, so that a pattern that merely spreads is not mistaken for one that inherits; the information an offspring carries about its partner is measured the same way. We then repeated the central experiment on a second, minimal machine whose instruction set we could edit, and proved what could be proved.

Here we show that life in these worlds begins in a fixed order. A sterile order of self-written zero bytes forms first. Then a two-byte word that loads itself into a register and pushes the register into memory becomes the first heritable replicator in almost every world; it needs no loop and never reads its surroundings, but its execution runs on from its last byte into its partner, and its offspring are mostly the partner's. Within a few hundred thousand interactions it is displaced, convergently and down to the byte, by a descendant carrying one control-flow cycle whose offspring are entirely its own. A theorem says why closure requires a cycle and why an instruction that writes its own operand forces the first replicator to be open;

a second machine confirms that adding such an instruction switches the open beginning on, and that the lethality of the order preceding life decides how the open phase ends. The first living thing is the simplest object its chemistry makes, no more complex than the sterile order around it, and structure-reading detectors fire on that order before life exists.

Sterile order precedes life

The soup holds 20,000 programs of L bytes on a 160×125 lattice (Fig. 1a). At each step 8,192 random cells are drawn and each is paired with a lattice neighbour; the pair is concatenated into a 2 L -byte ring, executed by a Z80 core for 128 instructions, and written back; a background mutation replaces a fixed number of random bytes per step, and nothing else happens: no fitness function, no replacement, no death (Methods). Heredity is measured by intervention. A tape is lifted out and executed against random partners; the fraction of partners that become copies is its copy success, and it is *heritable* if those copies copy in turn (Methods). We date emergence by the first heritable tape among the most common, and follow the heritable fraction of random cells as the population measure.

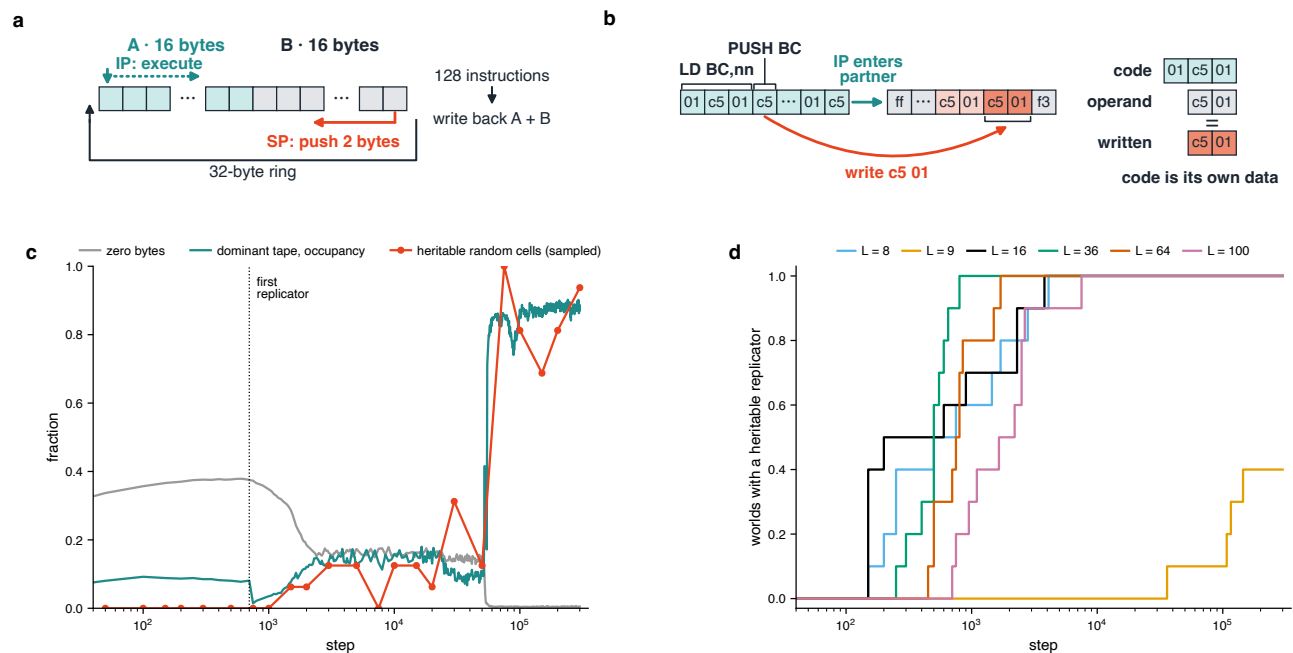


Fig. 1 | The soup and the order of events. **a**, An encounter: organism A (teal, 16 bytes) and partner B (grey, 16 bytes) share a 32-byte ring; the instruction pointer (IP, teal) starts at A's first byte and moves right, the stack pointer (SP, vermilion) starts at B's last byte and moves two bytes left with every push; after 128 instructions both halves are written back. Repeated cells are abbreviated. **b**, The first replicator, the word 01 c5 repeated: LD BC,nn loads its own next two bytes, c5 01, into BC, and PUSH BC writes exactly those two bytes at the stack pointer (vermilion; the lighter pair, the next push), so the partner fills from its far end with the organism's own word while the pointer runs on into the partner. Right, the instruction's bytes, its operand and what the push writes are the same word. **c**, One world ($L = 16$, seed 2001): fraction of zero bytes (the tar, grey), occupancy of the dominant tape (teal) and heritable fraction of 16 random cells at the sampled steps (vermilion, markers) against step; dotted line, first heritable replicator. **d**, Kaplan-Meier fraction of worlds with a heritable replicator by step for $L = 8-100$ (Stage E, 10 worlds per length, 300,000 steps).

In every unablated world the first thing to form is order that does not inherit. At the moment the most common pattern first occupies a tenth of the soup, between 24% and 34% of all bytes are zero at every tape length from 9 to 100 (Fig. 1c). The zeros are self-written: push, call and reset instructions executed with empty registers write zeros wherever the stack pointer points, and zero is itself an instruction that does nothing, so the pattern spreads by being written and costs nothing to execute; its carriers produce no copies that copy in turn. We call this flood *tar*, after the asphalt that prebiotic chemistry makes from the reactions that also make sugars¹³: order without heredity, made by the chemistry that will make life.

The tar is what detectors see. The high-order entropy used to detect the transition in random-program soups⁵ is anti-correlated with life in our worlds of 25 to 81 bytes (area under the curve 0.07–0.34 against the culture-test ground truth), because the flood compresses better than the replicators that replace it (Extended Data Fig. 5). A complexity-times-abundance measure of selection^{19,44} behaves the same way: it rises a median nineteen-fold above its random-soup value before the first heritable replicator exists in 71 of 80 worlds, and as a detector of heredity across tape lengths it is no better than compression (AUC 0.72 against 0.75; Extended Data Fig. 12). The reason is that the first living thing is not more complex than the tar: the assembly index of the first replicator is the lowest any string of its length can have, equal to that

of the modal sterile tape in 79 of 80 worlds. Order comes first, it is as simple as what follows, and only intervention tells the two apart.

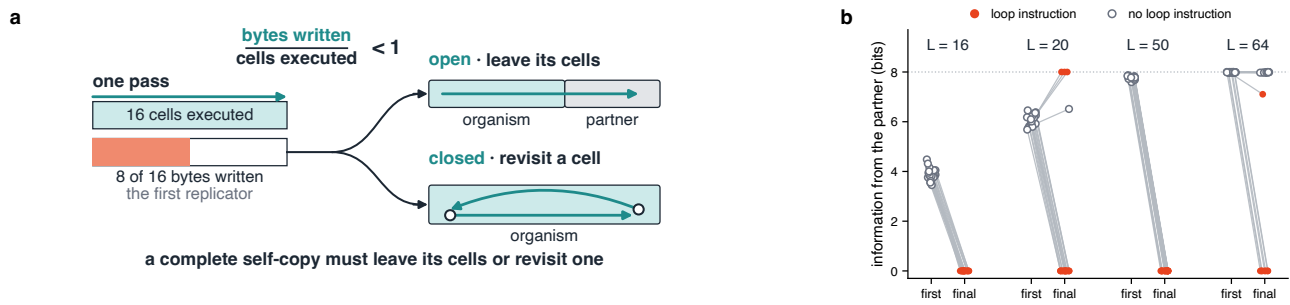


Fig. 5 | Closure as a cycle. **a**, Theorem 2 as a diagram. A program that writes fewer bytes than the cells it executes cannot copy itself in one pass: the first replicator ($L = 16$) executes its 16 cells and has written 8 of the 16 bytes a copy needs (executor trace), so copying requires more execution, which can only continue outside its cells (open) or revisit one of them (closed, a cycle). **b**, Information the offspring carries about the partner, $H(o | x)$ in bits over 256 random partners, for the first replicator and the final dominant of every Stage G world, by tape length (lines join the same world; filled vermilion, loop instruction present; open grey, absent; dotted line, the 8-bit ceiling of 256 partners). The closure-window model ($q \int n dt$) and the BFF variants are in Extended Data.

The first replicator is a literal

The first heritable replicator is the same program almost everywhere: the two-byte word `01 c5`, repeated to fill the tape (Fig. 1b). In the pre-registered confirmatory stage it is the modal first tape in 19 of 20, 12 of 20, 14 of 20 and 18 of 20 worlds at $L = 16, 20, 50$ and 64 , the rest being its register variants, at a Kaplan–Meier median of 450–750 steps whatever the length (Fig. 1d). Executed, the word is an instruction that loads the next two bytes into a register pair, followed by an instruction that writes that pair into memory at the stack pointer, which starts in the partner. The bytes the first instruction carries are the bytes the second writes: the program is its own data, and it copies itself as a side effect of being run. It has no loop, because the ring returns the pointer; no counter, because every byte of the tape is the same word; and no test of its surroundings, because it never reads them.

It is also nearly the only thing of its size that can replicate. Of all 65,536 two-byte words tiled to 16 bytes, exactly five copy themselves into an all-zero partner, all of them load-and-push pairs, and none copies into more than 81% of random partners (Extended Data Fig. 8). A further 246 words pass the heredity threshold without producing a single faithful copy, by smearing return addresses, which is why faithfulness as well as heredity is required before anything is called a replicator.

Openness limits the first replicator

The first replicator is *open*. Executed against 256 random partners it copies into a median 0.66–0.73 of them (range 0.56–0.84 over 80 worlds and four lengths) and loses a quarter or more of its own bytes in a median 0.06–0.43 of encounters (Fig. 2a,b). The failures have one cause: after writing its copy the pointer runs on into the partner and executes the partner’s bytes, which can poison the register pair before the next push. Against all-zero partners the pusher copies perfectly; its success falls as the density of active bytes in the partner rises, and no execution budget rescues it.

Openness has a measurable meaning. With the organism fixed and execution deterministic, the information that flows from the partner into the offspring equals the entropy of the offspring over random partners (Methods). For the first replicator that entropy is a median 7.0 bits of a possible 8, from 3.9 bits at $L = 16$ to 8.0 at $L = 64$, and in no world below 1 bit (Fig. 5b): most of what a child looks like is decided by the neighbour it was made against. The population shows it. At $L = 16$ the heritable fraction of random cells stays between 0.07 and 0.19 for the first 20,000 steps after emergence (Fig. 2c): the first replicator’s reign is a world of damaged copies.

Closure evolves as a control cycle

The first replicator does not stay. By 300,000 steps the dominant tape at $L = 16$ carries a control-flow instruction in 20 of 20 worlds, and 17 of them end on the identical eight-byte sequence whose conditional return sends the pointer back into the organism’s own body (Extended Data Fig. 2); at $L = 50$ all 20 worlds carry a relative jump inserted into the

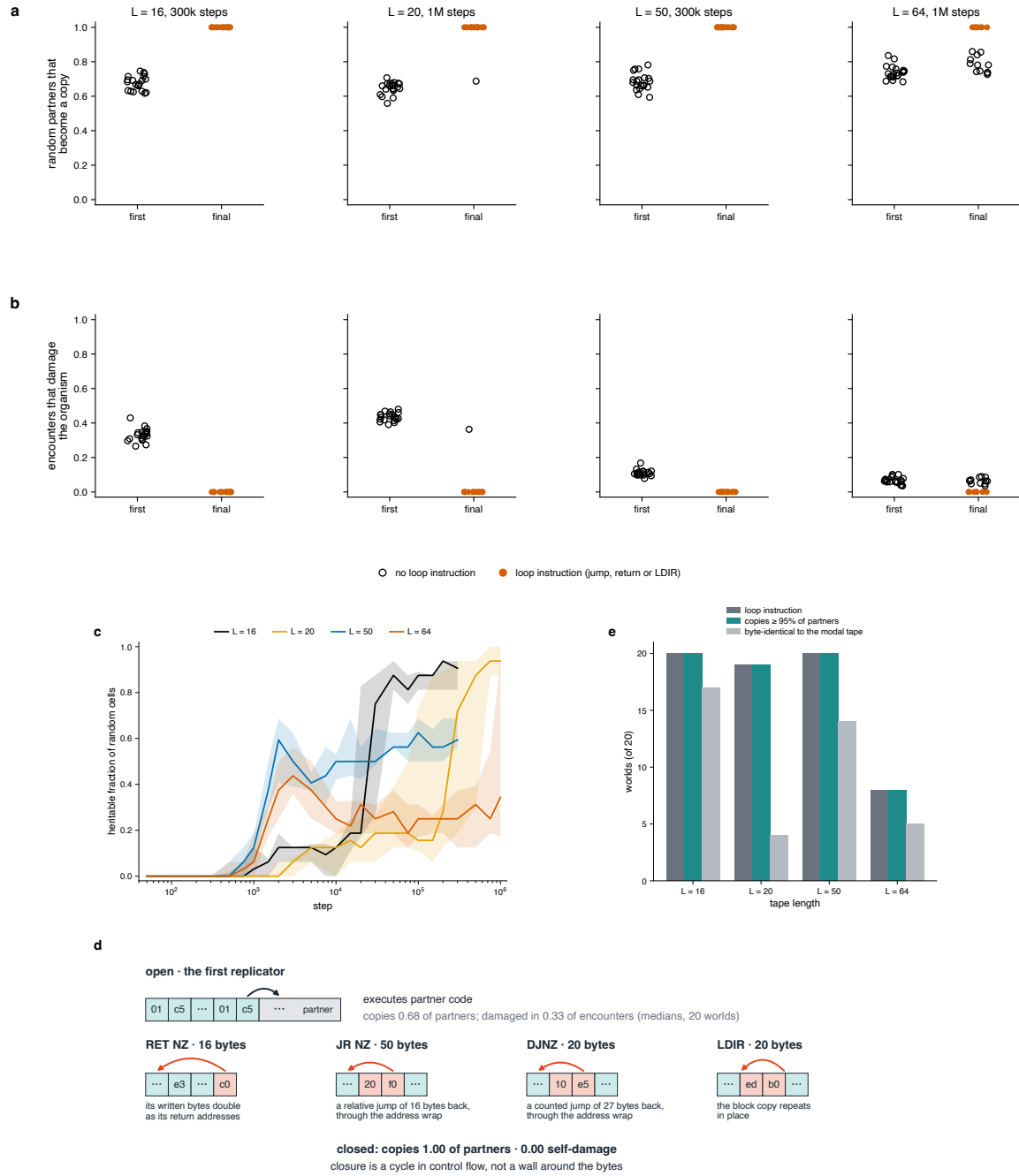


Fig. 2 | The first replicator is open, the successor closed. **a**, Fraction of 256 random partners that become a $\geq 75\%$ copy of the first heritable replicator and of the final dominant, one point per world (Stage G, 20 worlds per L; open circles, no loop instruction; filled, jump, return or LDIR). **b**, Fraction of encounters in which the organism loses $\geq 25\%$ of its bytes. **c**, Heritable fraction of 16 random cells against step (median and interquartile range over worlds). **d**, Control flow of the first replicator and of four closed successors, schematically (bytes abbreviated; the instruction that closes the cycle is shaded and its return drawn as a vermilion arc): the pusher's pointer runs on into the partner, whereas RET NZ returns to addresses formed from the organism's own written bytes, JR NZ and DJNZ jump back (through the 16-bit address wrap), and LDIR repeats in place. Medians of the partner test over Stage G worlds: the first replicator copies 0.68 of partners and is damaged in 0.33 of encounters; every closed successor copies 1.00 with 0.00 self-damage. Exact traces, Extended Data Fig. 2. **e**, Worlds (of 20) whose final dominant carries a loop instruction, copies $\geq 95\%$ of partners, and equals the modal final tape byte for byte.

pusher tiling, 14 of them byte-identical (Fig. 2e). Across the 233 unablated worlds of the exploratory stages, 92 gained a control-flow instruction between the first replicator and the final dominant and 3 lost one (exact McNemar $P = 7 \times 10^{-24}$). Every such successor copies into 1.00 of 256 random partners with 0.00 self-damage, whichever instruction implements the cycle: a conditional return, a relative or counted jump, an indirect jump, or the hardware block-copy loops (Fig. 2a,b,d). Its offspring carry nothing from the partner: the information inflow is exactly zero in 63 of the 67 worlds whose final dominant has such an instruction, and the four exceptions copy all but one to three bytes, which remain the partner's (Fig. 5b). The heritable fraction of random cells rises to 0.75–0.94 when the closed successor takes over (Fig. 2c).

We call this property *closure*: the organism's bytes are a forward-invariant set under execution in every context (Methods). It is operational closure¹ made countable and individuality⁴ made measurable by intervention. It is not a membrane, since the closed organism's bytes are as exposed to writes as before; not a test of the environment, since the return tests a flag the organism set itself; and not a property of place, since a pre-registered control with well-mixed pairing (ten worlds, $L = 16$) found the open phase exactly as heritable as on the lattice and closure later rather than sooner, reached in 7 of 10 worlds against 20 of 20, although the first replicator met its own kind in 57% of encounters on the lattice and 11% when mixed (Methods). It is a cycle in the organism's control flow that never leaves the organism (Fig. 2d). Closure takes longer the larger the organism: at $L = 20$, 19 of 20 worlds are closed by one million steps; at $L = 64$ only 8 of 20, with 12 still dominated by the open pusher, which there is also the least damaged open form in the atlas (self-damage 0.06) and so under the weakest pressure to close. The pre-registered expectation of at least 10 of 20 closures at both lengths was met at $L = 20$ only behaviourally, by a family that closes with the block-copy loop our syntactic criterion did not count, and not at $L = 64$ (Methods).

Three instruction families are load-bearing

Which instructions does this need? The seed-paired ablation atlas (Fig. 3a) finds three load-bearing parts and nothing else. Removing *immediate loads* leaves 2 of 10 worlds alive at 300,000 steps; removing *stack writers* delays emergence 160- to 992-fold in median and hands the soup to a replicator built on the block-copy instruction; removing *all copying* (112 opcodes) or leaving only read-modify-write instructions produces no replicator in 40 worlds at one million steps. Every other ablation, block copy, calls and returns, memory loads, input and output, exchanges, is within seed-paired noise of the unablated soup.

The size axis has no floor but a dead zone (Fig. 3b). Emergence is certain from $L = 16$ to $L = 100$ and its median time rises only eightfold over that range, because tiling makes a large organism of a two-byte word; below 16, life appears at $L = 5, 6$ and 8 but in only 4, 4 and 6 of 10 worlds at $L = 9, 10$ and 12 , exactly where the pusher's isolated heritability dips to 0.20–0.32. The zone is geometric: the stack pointer descends two bytes per push while the instruction pointer ascends, and when a push lands on the word the pointer is about to read, the register pair is poisoned. Padding the pair memory to a longer ring without changing the organism moves the collision and switches the zone off (Fig. 3c; Extended Data Fig. 3). Tar can prevent life as well as precede it: at $L = 9$ removing the stack writers makes life *more* likely (12 of 20 against 6 of 20 worlds at 128 steps, 19 of 20 against 5 of 20 at 512; Cochran–Mantel–Haenszel $P = 3.9 \times 10^{-6}$; Fig. 3d), because their tar destroys nascent block-copy replicators before they assemble: such a tape reaches the ten most common in 5 of 20 worlds at a median 118,500 steps with stack writers present and in 20 of 20 within 3,400 steps without. The execution budget decides whether an organism can be faithful. At $L = 100$ the block-copy regime is faithful in 1 of 10 worlds at 128 steps per encounter and 10 of 10 at 256, while at 64 steps no exemplar ever reaches 0.5% of the soup yet 6 of 10 final soups are heritable populations with no clone: heredity carried by partial copies, the communal regime placed before the Darwinian threshold¹⁴ (Extended Data Fig. 4).

One instruction decides how life begins

If the order of events is forced by the literal channel, a machine without one should be born closed, and giving it one should start life open. BFF, the Brainfuck family of Agüera y Arcas et al.⁵, has no literal channel: its instructions increment in place or copy from memory, and a one-pass execution cannot copy a 64-byte tape with three or more distinct bytes (Methods); an exhaustive search over 16.4 million structured straight-line programs found only one-symbol fills, the BFF analogue of tar. Eighty-four soups of 2^{17} programs run for 16,384 epochs test the prediction (Fig. 4). In BFF as published, 9 of 24 worlds produced a heritable replicator; every first replicator contains a loop (9 of 9) and none is open (0 of 9). A wrapping pointer produces more transitions (19 of 24; Fisher $P = 0.008$) but the same birth: 19 of 19 loop-bearing, 0 of 19 open. Adding one instruction that writes its own two operand bytes below the write head changes the beginning completely: in 12 of 12 worlds the first replicator is the one-byte tiling of that instruction, whose operands are two more copies of itself,

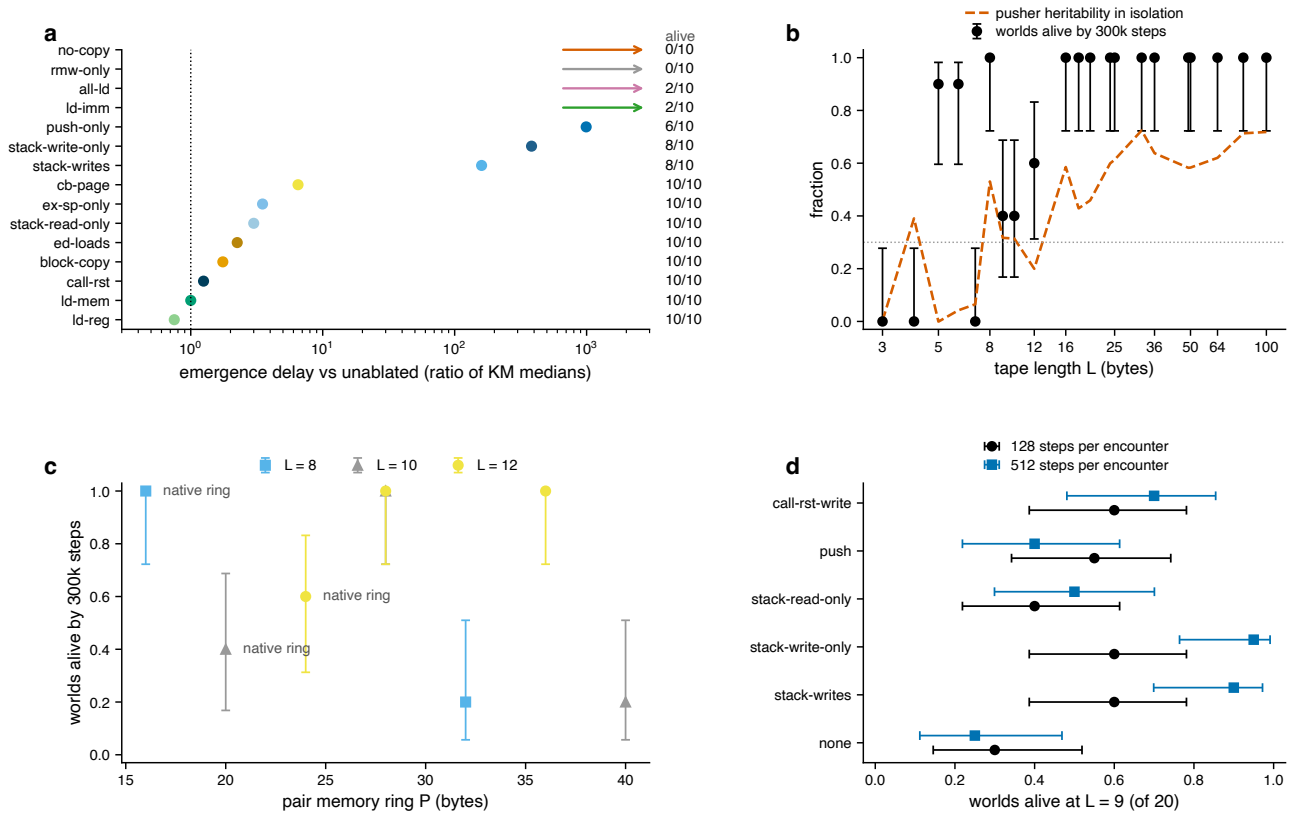


Fig. 3 | What the instruction set must provide. **a**, Delay of emergence relative to the unablated soup (ratio of Kaplan–Meier medians, 128 steps, mutation 1/16, seeds 101–110) for each ablation; arrows, no emergence; right margin, worlds alive by 300,000 steps. **b**, Worlds alive by 300,000 steps against tape length (Wilson 95% intervals) and the pusher’s heritability in isolation (dashed; dotted line, the 0.3 threshold). **c**, Worlds alive against pair-memory ring length P at L = 8, 10, 12 (Stage F; the native ring is 2L). **d**, Worlds alive at L = 9 (of 20) by ablation at 128 and 512 steps per encounter (Stage D).

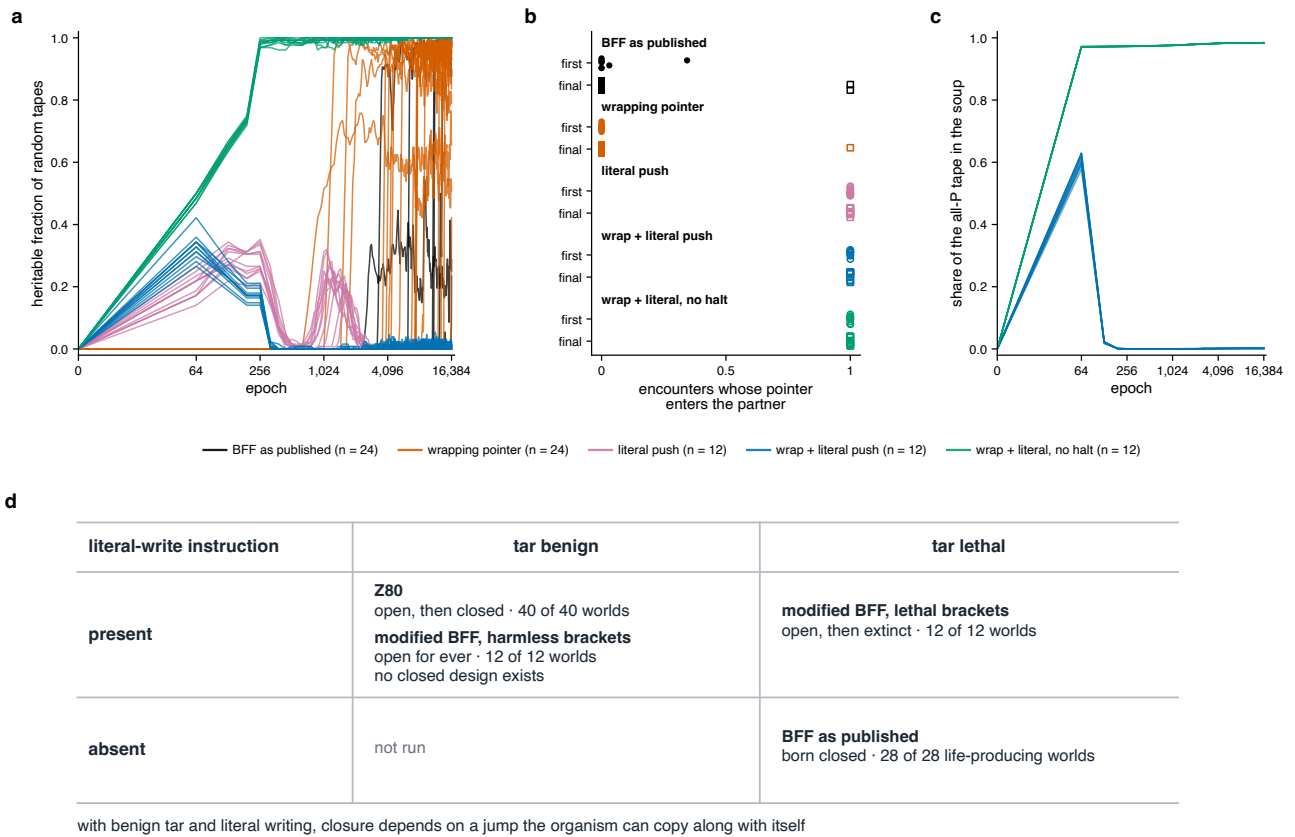


Fig. 4 | One instruction decides how life begins in BFF. **a**, Heritable fraction of 32 random tapes against epoch for every soup, by variant (as published, $n = 24$; wrapping pointer, $n = 24$; literal push, $n = 12$; wrap + literal push, $n = 12$; wrap + literal with unmatched brackets as no-ops, $n = 12$); the epoch axis is linear to 64 and logarithmic beyond, here and in **c**. **b**, Fraction of culture-test encounters in which the pointer enters the partner, for the first replicator (circles) and the final dominant (squares) of every life-producing soup, grouped by variant (open symbols, no loop instruction; filled, loop instruction). **c**, Share of the all-P class against epoch under wrap + literal (blue) and wrap + literal with no-op unmatched brackets (green), 12 worlds each (zero when the class is outside the ten largest): the same organism collapses when unmatched brackets halt the pointer and persists when they are no-ops. **d**, Classification of the substrates by two properties, a literal-write instruction (present or absent) and the lethality of the tar to an open replicator, with the observed order of events and world counts in each cell.

loop-free, with its pointer entering the partner in every encounter, present at the first sample with 53–60% of the soup. It is the 01 c5 of BFF, and its offspring carry 8 bits of the partner, the ceiling of the measurement (Extended Data Fig. 11).

Then it dies. Within about 200 epochs each of the 12 soups turns into tar that halts a pointer within a few dozen steps, since an unmatched bracket ends BFF execution; copies into such partners are cut short and sterile, and the open replicator persists only as a 0.2% minority, heritable in isolation and unable to spread. No closed class evolves and none can: no closed heritable replicator exists among 1.3 million periodic programs of this language up to period 10 (Methods). The open phase is a *window*, and one more switch shows what sets its length. Making an unmatched bracket a no-op instead of a halt, benign tar, leaves the same open replicator in place in 12 of 12 worlds for all 16,384 epochs with every random tape heritable at the end (Fig. 4c): a substrate in which life is open for ever. Open here means that execution enters the partner; by the information criterion this organism is already closed, because under a wrapping pointer its pushes lap the ring and overwrite everything, so its offspring is the same string in nearly every context (0.04 bits), which is why it never needs a cycle. Together these complete a classification (Fig. 4d): a literal channel decides whether life starts open or closed; the lethality of the tar decides whether an open beginning lasts; and the supply of copyable closers decides whether a lasting open phase ends in closure, as in the Z80, or never does.

Closure requires a cycle

Two counting statements make the order of events a theorem rather than an observation (Methods, Theorems). A straight-line program's pointer runs on into its partner, so a straight-line replicator is open by construction; and a replicator that writes fewer bytes than the cells it executes cannot complete a copy of itself in one pass over its own cells, so a closed replicator must execute some address twice: closure requires a cycle (Fig. 5a). A straight-line replicator exists exactly when the instruction set can write a literal, content independent of the memory being overwritten, at a self-advancing pointer; where it exists it is the size of the literal, strictly smaller than the smallest closed replicator, and present in a random soup with exponentially higher probability. The boundary of the theorem is itself a prediction, and the second machine confirmed it: a copier that writes at least as much as it reads may close by saturation without a cycle, which is what the wrapping literal pusher of BFF does.

In information terms closure is zero inflow from context to offspring, $I(o; e | x) = 0$, the limit that perfect fidelity across contexts requires (Methods, Lemma 7); selection for fidelity is selection against listening to the environment, and Fig. 5b is that statement measured, a median 7.0 bits falling to 0. What the theorems do not decide is the dynamics after the open beginning. Closure arrives with probability about $1 - \exp(-q \int n dt)$, where $\int n dt$ is the integral of the open population over time and q the rate of closing mutations; benign tar lets the integral grow until a closer is found, lethal tar closes the window first, and whether a copyable closer exists decides what a lasting window can become (Extended Data Fig. 11).

Discussion

The order of events is not a contingency of one machine. It follows from counting: whatever writes fewer symbols than it reads must, on its first pass over itself, either leave itself or return, and the smallest thing that writes itself has no room for a return. So the first replicator is open, and openness has a measurable meaning, the information its offspring carry about the context it copied into, which here is seven of eight bits. Closure is the return, and it too has a measurable meaning, zero bits, which four different instructions achieve in different worlds. The explanation has no adjustable part: the dichotomy is arithmetic, the literal channel is a property one reads off a substrate, and the lethality of by-products is a measurement. It reaches beyond its origin, to any sequential copying process with overhead, and it answers questions it was not built for: why the first replicators are sloppy and only partly heritable; why individuality can precede compartments; why a cycle, not a membrane, is the first thing selection builds; why complexity-based signatures of life must fail at the origin, the first living thing being no more complex than the sterile order that precedes it; and why the first innovations after replication should concern control, where copying starts, stops and returns, rather than function or code. The boundary of the rule is itself a prediction, and the second machine confirmed it: a copier that writes at least as much as it reads can close by saturation, overwriting everything it touches, and needs no return; the literal pusher of BFF under a wrapping pointer does exactly this, entering its partner in every encounter yet importing nothing from it.

These are predictions for chemistry, not results about it, and they can be tested. Template copying is a literal channel: a strand specifies its complement by physical fit, with no interpreter, so the first replicators in such a chemistry should be the shortest self-templating repeats, dependent on their surroundings and productive mainly of partial copies; the experiment is to measure heritable propagation among sloppy copies by transfer into fresh material, and openness as the variation of the product with its context, rather than to search for a clean self-replicating species. By-products are a variable, not a

nuisance: whether a chemistry's tar is inert or inhibitory to the copying process should decide whether an open phase lasts long enough for control to evolve, which a screen of nonenzymatic systems with matched inert and inhibitory additives can test¹³. Processivity is the chemical write ratio: vertical lineages should appear exactly when one copying event spans the genome, and below that heredity should be carried by the population without clones, as our 64-step soups show and as polymerase ribozymes of increasing processivity could test⁴⁵. In laboratory evolution of RNA replicators the earliest fixed changes should be of the control kind, circularisation, terminators, primers⁴². For the search for life elsewhere the lesson is methodological: abundant low-complexity order precedes and surrounds the first replicators and trips compression and assembly measures alike^{19,44}, so detection should be by intervention, adding fresh substrate and looking for propagation that depends on the inoculum, and order without heredity should be sought as a precursor signature. What limits the claims is plain. One real instruction set and one toy language were studied; the culture test is our own ground truth, though the order of events is reproduced under every setting we tried; closure at the largest tape lengths is in progress rather than complete within our horizons; and interactions in our soup are local. A pre-registered well-mixed control (ten worlds, $L = 16$) found the open phase exactly as heritable as on the lattice and closure later rather than sooner, reached in 7 of 10 worlds by 300,000 steps against 20 of 20, so the order of events does not depend on spatial structure; the first replicator meets its own kind in 57% of encounters on the lattice and 11% when mixed, and what space seems to change is not whether closure comes but how reliably⁴¹. Within those limits the result stands: heredity came first, individuality was a return in control flow, and the step that made a replicator an organism was the step that made it stop listening to its environment.

Methods

Soup. The soup holds $N = 20,000$ tapes of L bytes (L from 3 to 100), initialised with uniformly random bytes, on a 160×125 square lattice with local interaction. At every step 8,192 cells are drawn uniformly at random and each is paired with one of its four lattice neighbours (a step off the edge is reflected; the lattice does not wrap); a pair is active if neither member is already in another pair that step, which leaves $\approx 3,980$ active pairs (0.395 active interactions per cell per step; the count varies by a few per cent with concurrent GPU load and does not affect results). An active pair is concatenated, organism first, into a pair memory of P bytes ($P = 2L$ except in the padding experiment of Stage F, where $P \geq 2L$ and the padding bytes are never written back), executed by the Zilion Z80 core (@neovand/zilion 0.2.0, WGS) for a fixed budget of 128 instructions (32, 64, 256, 512, 1,024 or 2,048 in the budget arms) from byte 0 with all registers zero and the stack pointer at the last byte of the partner, and both tapes are written back. After each step $8,192/2^k$ bytes chosen uniformly at random over the soup are replaced by random bytes ($k = 4$ unless stated; $k = 1-6$ in the mutation arms). There is no fitness function, no replacement and no death. Instruction-set ablations replace whole opcode families by no-ops through a prefix-aware fetch hook that sees the operation after DD/FD and ED/CB prefix resolution; the thirteen ablations and their opcode counts are listed in Extended Data Table 2 (none, block-copy (4 opcodes), stack-writes (46), stack-write-only (22), stack-read-only (24), push-only, ex-sp-only, call-rst, call-rst-write, ld-mem, ld-imm, ld-reg, all-ld, ed-loads, cb-page, no-copy (112), rmw-only). The emulator was validated against an independent Z80 reference with zero genuine divergences in randomised 5,000-pair tests with and without ablation; it is not treated as ground truth for the Z80 itself, and no claim depends on cycle-exact timing.

Recording. Every run writes a sample at fixed steps (every step for the first 30, then a geometric schedule, then every 500 or 1,000) with: the ten most common exact tapes and their shares; quasispecies occupancy (fraction of cells within Hamming distance $\lfloor L/4 \rfloor$ of the dominant tape); byte histogram, zero-byte fraction and byte entropy; brotli-compressed bits per byte (quality 2) and the high-order entropy of ref. 5; species entropy and unique-tape fraction; per-interaction statistics (bytes written into the partner, copy events, destroyed programs); 16 random tapes for the functional census; and full soup snapshots at fixed steps and at emergence.

Culture test and partner tests. A tape runs as the first program against 64 uniformly random partners for the condition's budget; copy success is the fraction of partners that become $\geq 75\%$ copies at the best cyclic shift; gen2 is the copy success of the produced copies against fresh partners, weighted by the copy fraction; heritable means $\text{gen2} \geq 0.3$, faithful additionally $\geq 50\%$ of partners copied. Partner-independence tests use 256 random partners and record self-damage ($\geq 25\%$ of the organism's bytes changed in one encounter). In BFF the culture test also records whether the instruction pointer entered the partner in each encounter; closed means $\leq 5\%$ of encounters entered and $\geq 95\%$ of partners copied, open means $\geq 50\%$ entered. Information inflow from the environment, $I(o; e | x) = H(o | x) - H(o)$ for a deterministic executor, is the plug-in entropy in bits of the offspring (the partner's bytes after the encounter) over 256 random partners, ceiling 8 bits (results/biology/individuality/).

Events and statistics. t_{rep} is the first sampled step at which one of the three most common tapes has a share $\geq 0.5\%$ and is heritable by the culture test (a clonality criterion by design: it dates the first *clone*, not the first heritable material);

t_{faith} additionally requires faithfulness; $t_{q_{10}}$ (quasispecies occupancy $\geq 10\%$) is the pre-registered occupancy event and is reported because it fires on zero floods as well as on replicators. Emergence fractions are quoted as worlds reaching the event within the horizon with Wilson 95% intervals; times as Kaplan–Meier medians with censoring at the horizon. Ablation effects are ratios of Kaplan–Meier medians with seed-paired sign tests (the same seeds run under every ablation); confirmatory contrasts use one-sided Fisher exact tests and the Cochran–Mantel–Haenszel test pooled over budgets; the paired gain of control flow uses the exact McNemar test; between-variant transition rates in BFF use two-sided Fisher tests and Mann–Whitney tests on event times with non-transitioning runs censored at the horizon. In BFF, where replicator populations are quasispecies with small exact-class shares, the “first replicator” is the most common tape class (a tape and its reverse count as one class) at the first sample at which it is heritable, and the population event is a heritable fraction ≥ 0.5 of 32 random tapes; t_{rep} is reported alongside. All thresholds (0.75 copy, $\text{gen2} \geq 0.3$, 0.5% share, 10% occupancy, 95%/5% closure) were fixed before the confirmatory stages. Same-seed Z80 runs diverge because GPU scheduling makes the pairing order non-deterministic, so every Z80 claim is a population claim over seeds (ten same-seed repeats spread as widely as ten seeds); the BFF soup is bitwise reproducible across machines (identical epoch statistics on an Apple GPU and an NVIDIA L40S for six seeds over 16,384 epochs).

Definitions and theorems. A *self-writer* is a tape fixed under its own execution in some context. *Closure*: the organism’s bytes are forward-invariant under execution for every context in the class of uniformly random partners. Theorem 1: no straight-line execution in one-pass BFF copies a 64-byte tape with three or more distinct byte values (writes + writer-head moves + source changes exceed 128). Theorem 2: if bytes written per byte of code executed is below one, a closed full replicator executes some address twice. Theorem 6: for any machine in which the pointer runs on into adjacent memory absent a control transfer and Theorem 2’s bandwidth condition holds, a self-consistent literal-write instruction implies a straight-line (open) replicator the size of the literal, and its absence implies every replicator is loop-bearing. Lemma 7: $I(o; e | x) > 0$ implies imperfect fidelity across contexts; perfect fidelity implies $I = 0$. Proposition 3 (computer-assisted): the two-byte fixed points of the Z80 against the zero partner are exactly the five load–push words. Full statements and proofs in Supplementary Information.

BFF. cubff default semantics verified from source: two 64-byte tapes concatenated, pointer and heads at byte 0, ten instructions, every other byte a no-op, unmatched brackets halt, 2^{13} steps per encounter, 2^{17} programs, random ordered pairs each epoch, mutation 2^{-12} per byte per epoch. Switches: *ip_wrap* (pointer wraps modulo 128); *literal* (byte 0x50 = $P \ a \ b$: write a at $\text{head1}-1$ and b at $\text{head1}-2$, then $\text{head1} -= 2$, skip the operands); *nohalt* (unmatched brackets are no-ops). Our GPU kernel reproduces cubff’s validation replicator and matches a CPU reference on randomised tests. 24 seeds each for the published and wrapping variants, 12 for each literal variant; events every 64 epochs; high-order entropy with brotli quality 2 as in ref. 5.

Searches. Two-byte census: all 65,536 words tiled to 16 bytes, culture test (32 partners) and partner test (256), plus the deterministic fixed-point test against the zero partner. BFF straight-line search: 16,447,860 programs (prefix ≤ 3 instructions + tiling unit of period 2–5 with a copy instruction), 2 random partners, wrapping pointer, alphabet prefilter then exact similarity, full culture test for survivors. Closed-replicator search: all 1,309,528 periodic tilings of $\{P, [,], \text{no-op}\}$ up to period 10 under the literal, wrapping, no-halt rules, three partners each.

Pre-registration. Stages D, F and G and the BFF cells were pre-registered with predictions and kill criteria before launch; every amendment is dated in the change log. Misses reported: the heritable-fraction thresholds of G1(c) failed at $L = 50$; the syntactic loop criterion of G2 omitted hardware loops (6/20 and 3/20 syntactic against 19/20 and 8/20 behavioural); the mechanism first claimed for an open BFF replicator under a wrapping pointer was falsified by the harness test before any soup ran and the prediction restated with both readings; a held prediction for the non-wrapping literal cell was corrected by the counting argument before it was run.

Experimental stages. Stage A (630 runs, $L = 16$): seven ablations $\times$ budgets $\{32, 128, 512\} \times$ mutation $\{1/4, 1/16, 1/64\}$, ten seeds, 300,000 steps. Stage B (640): tape lengths 4–100 under two ablations and three budgets. Stage C (490): thirteen ablations at $L = 16$, seeds 101–110, with one-million-step nulls. Stage D (280, confirmatory): the $L = 9$ reversal with seeds 1001–1020. Stage E (1,230): the size axis under four control arms, budget and mutation sweeps at $L = 100$, variance and exception cells. Stage F (210, confirmatory): padded pair memories. Stage G (80, confirmatory): closure with seeds 2001–2020 at $L = 16$ and 50 (300,000 steps) and $L = 20$ and 64 (1,000,000 steps). BFF: 24 seeds each for the published and wrapping variants, 12 for each of the three literal variants. Stage H (10, control): well-mixed pairing, the partner drawn uniformly from the soup, at $L = 16$, seeds 3001–3010, run locally. Post hoc analyses (closure, two-byte census, detectors, invasion, census dynamics, ring sweep) used recorded data and local GPU time only.

Compute, code and data. 3,560 Z80 runs and 84 BFF soups on Modal L40S GPUs plus local runs on an Apple GPU; $\approx$ \$400 at list price. The source code (simulator, assays, analysis and figure scripts), every condition file, every raw run summary, every generated table and the pre-registration documents with their dated change log are in the project repository

(to be archived with a DOI on acceptance); each figure is produced from the generated tables by one script, and each number in the text is traceable to a named table.

Extended Data

Extended Data Fig. 1 | Emergence by tape length and ablation. Kaplan–Meier fraction of worlds with a heritable replicator against step for every tape length (4–100) under the four Stage E control arms (nominal; constant mutation per byte; constant mutation per cell; constant budget per byte) and under the stack-writer ablation; 10 worlds per cell, 300,000 steps. Source: `results/stageE/stage_e/size_arms.csv`, `results/stageE/assays.csv`.

Extended Data Fig. 2 | The replicator zoo. First heritable replicator and final dominant tape of every Stage G world (80 worlds), as hex and disassembly, grouped by tape length, with copy success and self-damage from the partner test. Source: `results/stageG/ZOO.md`, `results/stageG/stageG/stage_g_runs.csv`.

Extended Data Fig. 3 | Ring geometry. Worlds alive and median emergence step for every tested (tape length, ring length) pair of the ring sweep (52 cells) and Stage F; the pre-registered ring-arithmetic law (period divides the ring) against the observed law (period divides the pair length 2L); the stack-pointer/instruction-pointer race that produces the dead zone. Source: `results/ring_sweep/ring_sweep.csv`, `results/stageF/stage_f/rings.csv`, `results/stageF/stage_f/pusher_trace.csv`.

Extended Data Fig. 4 | Budget and mutation at L = 100. Heritable and faithful worlds against steps per encounter (32–2,048) under the pusher and LDIR regimes; the sub-clonal regime at 64 steps (heritable fraction of random cells against dominant-tape share); median period of the first replicator and final high-order entropy against per-byte mutation rate (the error-threshold curve). Source: `results/stageE/stage_e/budget_L100.csv`, `mutation_L100.csv`.

Extended Data Fig. 5 | Life detectors against the culture test. Area under the ROC curve for each recorded statistic (high-order entropy, non-unique tape fraction, bits per byte, species entropy, interaction-based statistics, dominant-tape share, quasispecies occupancy, motif share) against the heritable fraction of random cells, by tape length and by step; the inversion of high-order entropy under the zero flood. Source: `results/detectors/`.

Extended Data Fig. 6 | Tar prevents assembly. Census of LDIR-bearing tapes in the ten most common tapes before emergence at L = 9, with and without stack writers (20 worlds per arm): first entry, episodes, destruction rate and zero bytes written per interaction. Source: `results/stageD/census/`.

Extended Data Fig. 7 | Invasion assays. The pusher and the LDIR unit seeded into 1% of cells at 32 and 128 steps per encounter and two mutation rates: occupancy against step, 20 worlds per cell. Source: `results/invasion/invasion.csv`.

Extended Data Fig. 8 | Every two-byte word. Culture-test score and gen2 for all 65,536 two-byte words tiled to 16 bytes; the five fixed points against the zero partner; the 246 return-address smears. Source: `results/census2/`.

Extended Data Fig. 9 | Searches in BFF. The 16.4-million-program straight-line search under a wrapping pointer (copy fraction and offspring byte diversity of the 130 programs passing the byte-match criterion) and the 1.3-million-tiling closed-replicator search under the benign-tar rule (open replicators by period; no closed heritable tiling). Source: `results/bff_search/`, `results/bff_closed_search/`.

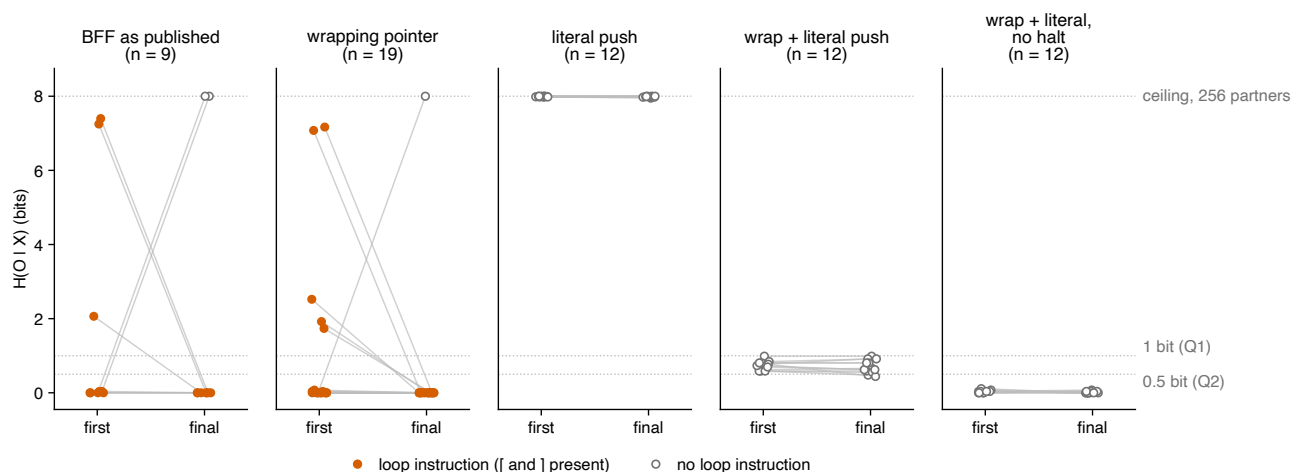
Extended Data Fig. 10 | Reproducibility and compute. Spread of emergence times across ten repeats of one seed against ten seeds in the Z80 soup (GPU scheduling nondeterminism); identity of epoch statistics between local Apple-GPU and Modal L40S runs of six BFF seeds over 16,384 epochs; GPU hours and cost per stage. Source: `results/stageE/stage_e/NUMBERS_E.md` (E6), `results/bff/reproducibility_local_vs_modal.csv`, `PLAN.md` change log.

Extended Data Table 1 | Final-tape classes per tape length (Stage G). Control-flow and block-repeat content, partner-copy fraction and self-damage of the final dominant, number of worlds per class. Source: `results/stageG/stageG/NUMBERS_G.md`.

Extended Data Table 2 | Ablation sets. Opcode families removed in each ablation, with opcode counts and example mnemonics. Source: `make_conds.ABLATIONS`, `src/lib/z80-opcodes.ts`.

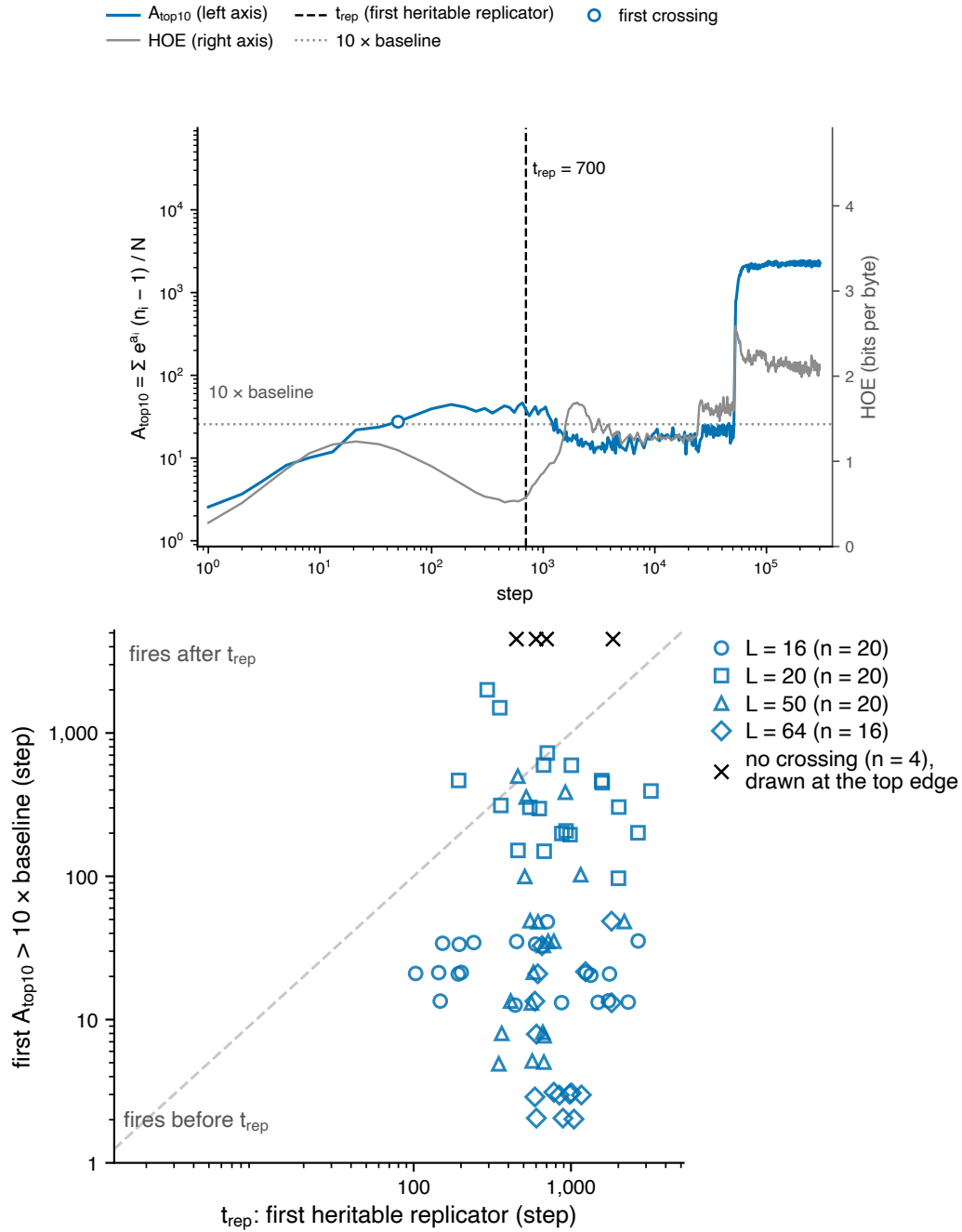
References (main text; verified entries from LITERATURE.md)

1. Maturana, H. R. & Varela, F. J. *Autopoiesis and Cognition* (Reidel, 1980).
2. Rosen, R. *Life Itself* (Columbia Univ. Press, 1991).
3. Fontana, W. & Buss, L. W. "The arrival of the fittest": toward a theory of biological organization. *Bull. Math. Biol.* **56**, 1–64 (1994).



Extended Data Fig. 11 | Information inflow in the second machine. $H(o | x)$ over 256 random partners for the first replicator and the final dominant of every life-producing BFF soup, by variant (lines join the same soup; filled, both brackets present), with the 8-bit ceiling of 256 partners and the pre-registered 1-bit and 0.5-bit thresholds. The published variant is born with near-zero inflow; the literal-push variant without a wrapping pointer sits at the ceiling; the two wrapping literal variants flood the ring and have near-zero inflow although their pointer enters the partner in every encounter. Source: [results/biology/individuality/](#).

4. Krakauer, D., Bertschinger, N., Olbrich, E., Flack, J. C. & Ay, N. The information theory of individuality. *Theory Biosci.* **139**, 209–223 (2020).
5. Agüera y Arcas, B. et al. Computational life: how well-formed, self-replicating programs emerge from simple interaction. Preprint at arXiv:2406.19108 (2024).
6. Rasmussen, S., Knudsen, C., Feldberg, R. & Hindsholm, M. The Coreworld: emergence and evolution of cooperative structures in a computational chemistry. *Physica D* **42**, 111–134 (1990).
7. Koza, J. R. Spontaneous emergence of self-replicating and evolutionarily self-improving computer programs. In *Artificial Life III* 225–262 (Addison-Wesley, 1994).
8. Dyson, F. *Origins of Life* 2nd edn (Cambridge Univ. Press, 1999).
9. Szostak, J. W. The eightfold path to non-enzymatic RNA replication. *J. Syst. Chem.* **3**, 2 (2012).
10. Kauffman, S. A. Autocatalytic sets of proteins. *J. Theor. Biol.* **119**, 1–24 (1986).
11. Maynard Smith, J. & Szathmáry, E. *The Major Transitions in Evolution* (Oxford Univ. Press, 1995).
12. Cicala, F. et al. Coevolution of self-replication and function in a digital primordial soup. Preprint at arXiv:2607.09211 (2026).
13. Benner, S. A., Kim, H.-J. & Carrigan, M. A. Asphalt, water, and the prebiotic synthesis of ribose, ribonucleosides, and RNA. *Acc. Chem. Res.* **45**, 2025–2034 (2012).
14. Woese, C. R. On the evolution of cells. *Proc. Natl Acad. Sci. USA* **99**, 8742–8747 (2002).
15. Eigen, M. Selforganization of matter and the evolution of biological macromolecules. *Naturwissenschaften* **58**, 465–523 (1971).
16. Eigen, M. & Schuster, P. The hypercycle: a principle of natural self-organization. Part A. *Naturwissenschaften* **64**, 541–565 (1977).
17. Nowak, M. A. & Ohtsuki, H. Prevolutionary dynamics and the origin of evolution. *Proc. Natl Acad. Sci. USA* **105**, 14924–14927 (2008).
18. Walker, S. I. & Davies, P. C. W. The algorithmic origins of life. *J. R. Soc. Interface* **10**, 20120869 (2013).
19. Sharma, A. et al. Assembly theory explains and quantifies selection and evolution. *Nature* **622**, 321–328 (2023).
20. Ray, T. S. An approach to the synthesis of life. In *Artificial Life II* 371–408 (Addison-Wesley, 1991).
21. Ofria, C. & Wilke, C. O. Avida: a software platform for research in computational evolutionary biology. *Artif. Life* **10**, 191–229 (2004).
22. Langton, C. G. Self-reproduction in cellular automata. *Physica D* **10**, 135–144 (1984).
23. Sayama, H. A new structurally dissolvable self-reproducing loop evolving in a simple cellular automata space. *Artif. Life* **5**, 343–365 (1999).
24. Hutton, T. J. Evolvable self-replicating molecules in an artificial chemistry. *Artif. Life* **8**, 341–356 (2002).
25. Dittrich, P., Ziegler, J. & Banzhaf, W. Artificial chemistries — a review. *Artif. Life* **7**, 225–275 (2001).
26. Sipper, M. Fifty years of research on self-replication: an overview. *Artif. Life* **4**, 237–257 (1998).



Extended Data Fig. 12 | A complexity-times-abundance measure against the culture test. The assembly measure over the ten most common classes (Re-Pair upper bound on the assembly index; Sharma et al.'s formula) and the high-order entropy against step for the exemplar world ($L = 16$, seed 2001) with the first heritable replicator marked; the step of the measure's first tenfold rise against the first heritable replicator for all 80 Stage G worlds; area under the ROC curve of both statistics against the heredity event by tape length (Stage E). Source: [results/biology/assembly/](#).

27. von Neumann, J. *Theory of Self-Reproducing Automata* (ed. Burks, A. W.) (Univ. Illinois Press, 1966).
28. Chaitin, G. J. To a mathematical definition of "life". *ACM SIGACT News* **4**, 12–18 (1970).
29. Szathmáry, E. The evolution of replicators. *Phil. Trans. R. Soc. Lond. B* **355**, 1669–1676 (2000).
30. Hordijk, W. & Steel, M. Detecting autocatalytic, self-sustaining sets in chemical reaction systems. *J. Theor. Biol.* **227**, 451–461 (2004).
31. Lincoln, T. A. & Joyce, G. F. Self-sustained replication of an RNA enzyme. *Science* **323**, 1229–1232 (2009).
32. Vaidya, N. et al. Spontaneous network formation among cooperative RNA replicators. *Nature* **491**, 72–77 (2012).
33. Moreno, A. & Mossio, M. *Biological Autonomy: A Philosophical and Theoretical Enquiry* (Springer, 2015).
34. Waddington, C. H. Canalization of development and the inheritance of acquired characters. *Nature* **150**, 563–565 (1942).
35. Wagner, A. *Robustness and Evolvability in Living Systems* (Princeton Univ. Press, 2005).
36. Adami, C. & LaBar, T. From entropy to information: biased typewriters and the origin of life. In *From Matter to Life: Information and Causality* (eds Walker, S. I., Davies, P. C. W. & Ellis, G. F. R.) 130–154 (Cambridge Univ. Press, 2017).
37. Knierim, P. et al. BFF: simple explanations for complex phenomena. Preprint at arXiv:2607.01483 (2026).
38. England, J. L. Statistical physics of self-replication. *J. Chem. Phys.* **139**, 121923 (2013).
39. Wong, B. Points of view: color blindness. *Nat. Methods* **8**, 441 (2011).
40. Pargellis, A. N. The spontaneous generation of digital "Life". *Physica D* **91**, 86–96 (1996).
41. Boerlijst, M. C. & Hogeweg, P. Spiral wave structure in pre-biotic evolution: hypercycles stable against parasites. *Physica D* **48**, 17–28 (1991). [added 2026-10-08; verify in LITERATURE.md]
42. Mizuuchi, R., Furubayashi, T. & Ichihashi, N. Evolutionary transition from a single RNA replicator to a multiple replicator network. *Nat. Commun.* **13**, 1460 (2022). [added 2026-10-08; verify]
43. (reserved)
44. Marshall, S. M. et al. Identifying molecules as biosignatures with assembly theory and mass spectrometry. *Nat. Commun.* **12**, 3033 (2021). [added 2026-10-08; verify]
45. Cojocaru, R. & Unrau, P. J. Processive RNA polymerization and promoter recognition in an RNA World. *Science* **371**, 1225–1232 (2021). [added 2026-10-08; verify]

All reference details above were checked against a source on 2026-10-08 (LITERATURE.md); DOIs to be added at formatting.